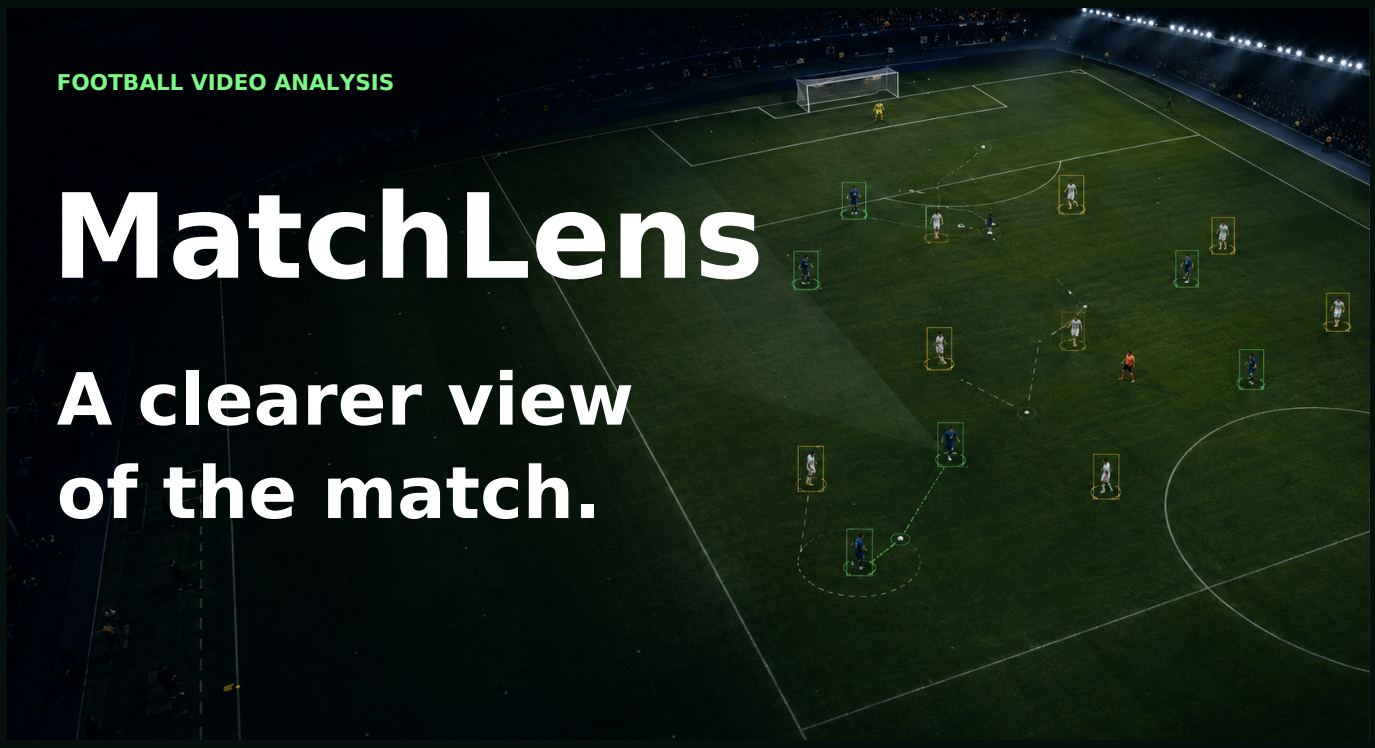


FOOTBALL VIDEO ANALYSIS

MatchLens

A clearer view of the match.

FOR CLUBS, COACHES AND ANALYSTS

MatchLens turns football footage into player tracking, possession insights, match events and performance metrics. Coaches and analysts can review the evidence, freeze the action and draw on the frame to explain a tactical idea.

TECHNICAL STACK

Inside MatchLens

VISION & ANALYSIS

Python / OpenCV

YOLO detection and ByteTrack tracking.

APPLICATION

React / Node.js

Interactive interface and engine launcher.

LIVE DELIVERY

WebSockets / MJPEG

Analysis updates and video streaming.

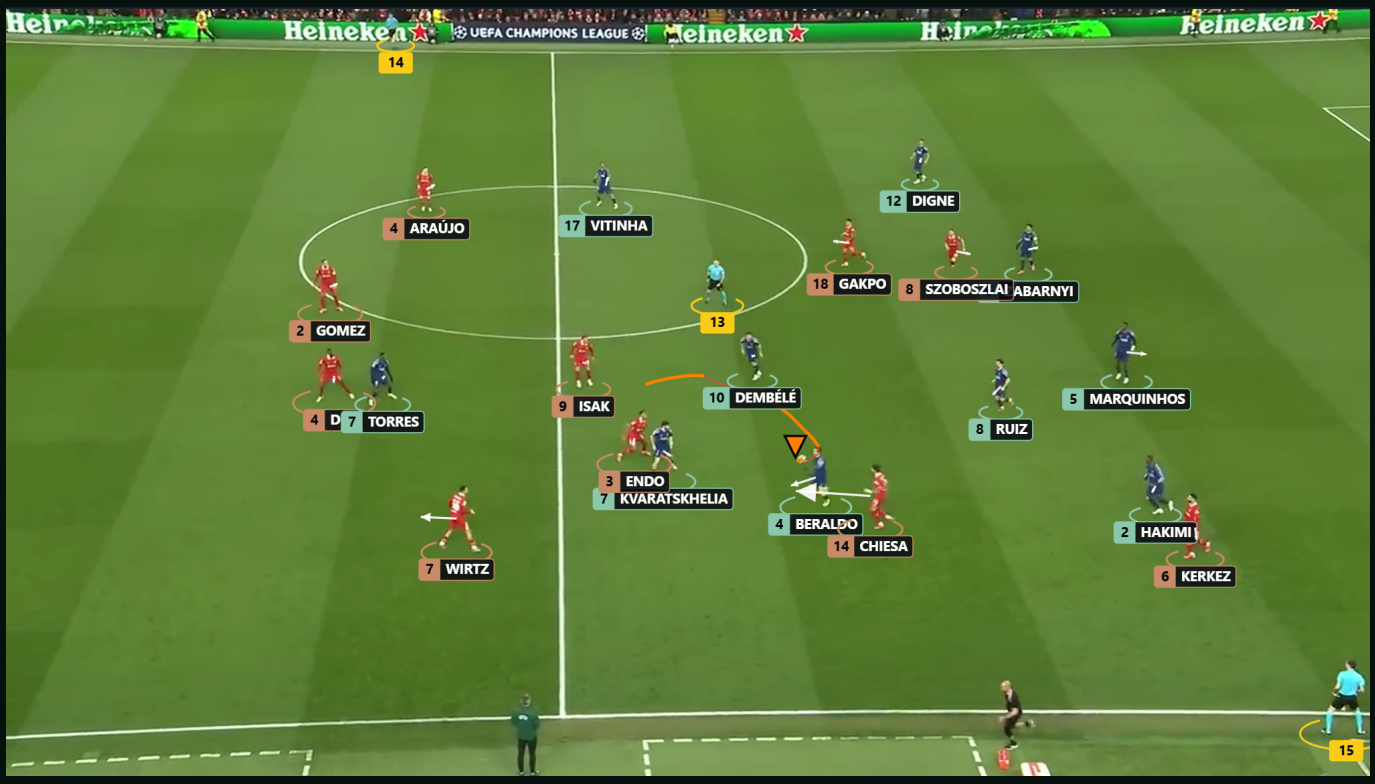
MATCH DATA

Parquet

Structured event records for later analysis.

What MatchLens reveals

Explore the same match through movement, control and meaningful football actions.



Project output: player tracks, team colours and name labels

01 Tracking & identity

Follow players and the ball over time. Connect tracks to teams and player names using formation and position cues.

02 Possession

Understand who controls the ball and how control changes between players and teams.

03 Match events

Identify passes, carries, crosses, shots and restarts. Turn continuous play into moments that can be reviewed.

04 Performance maps

Explore where players operate and where actions occur. Add spatial context to match and player analysis.

OBSERVE

Players and ball

UNDERSTAND

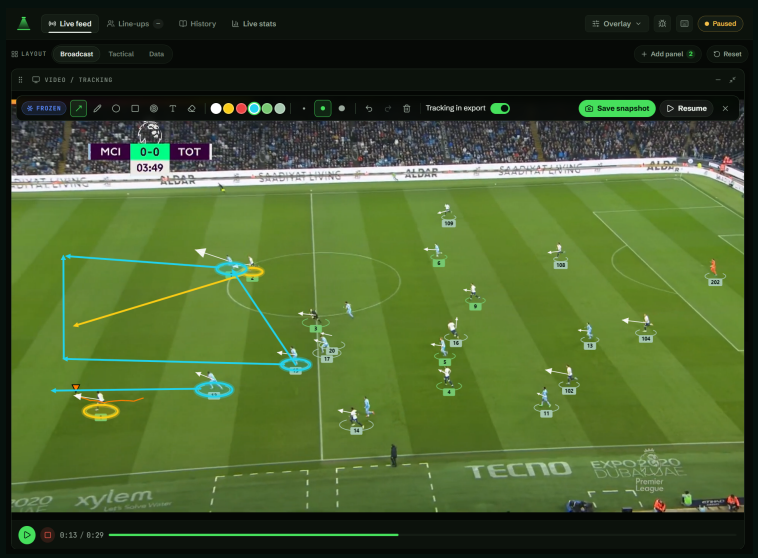
Positions and control

ANALYSE

Events and value

Review. Draw. Explain.

Freeze a moment during playback, draw over the frame and use match statistics to support your tactical explanation.



MatchLens overlay: paused frame with tactical drawings

Tactical overlays

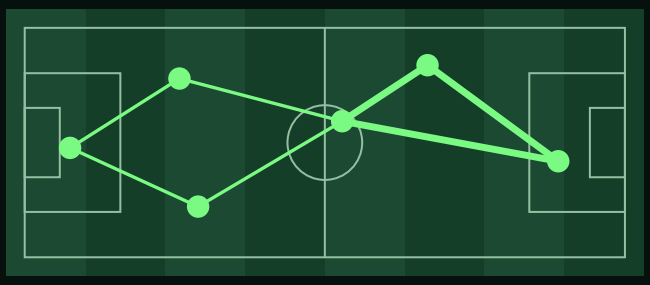
Pause on the moment that matters. Draw on the frame to explain positioning, movement or a tactical adjustment.

Review statistics and tactical insights alongside the action.

FOUR VIEWS FOR DEEPER ANALYSIS

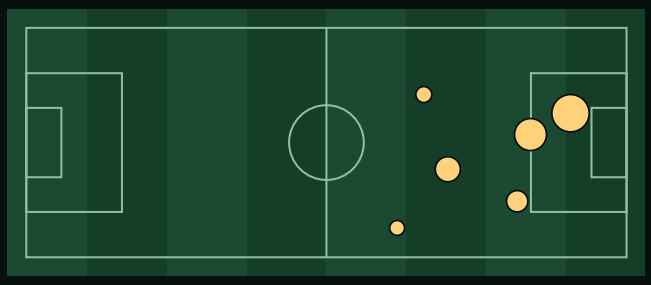
ILLUSTRATIVE EXAMPLES

Passing map



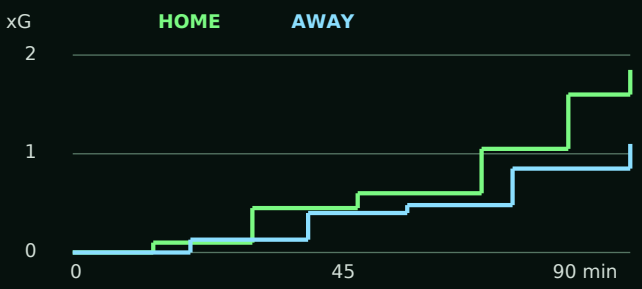
See passing connections and the routes a team uses to build attacks.

Shooting map



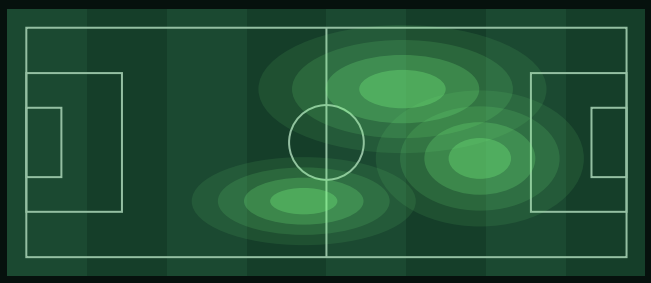
Locate attempts and compare chance quality. Here, larger dots mean higher xG.

xG race



Compare how each team accumulates expected goals over match time.

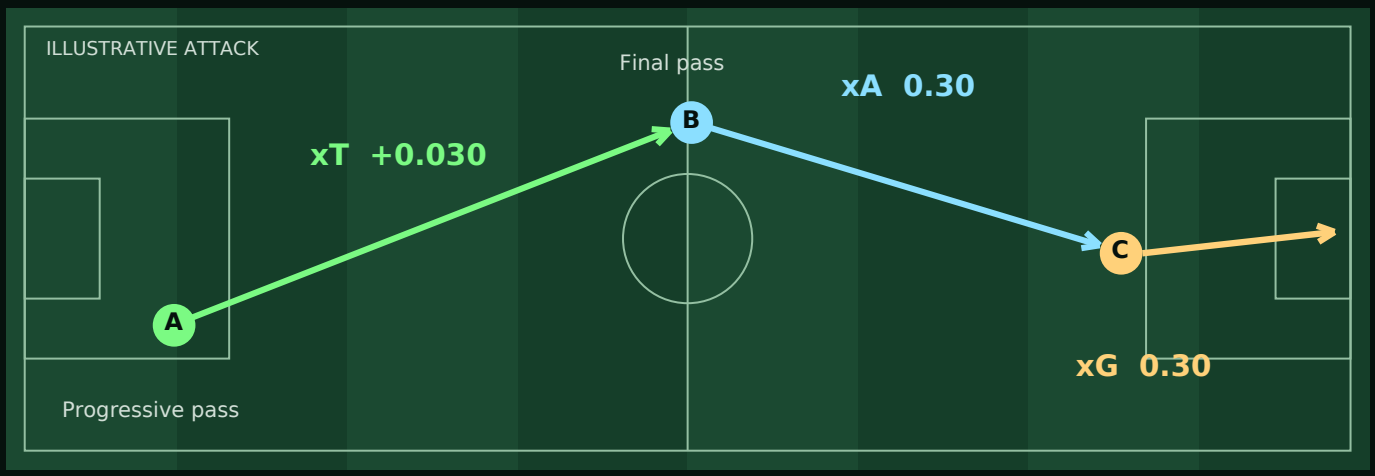
Heat map



Reveal the areas where a player or team is most active.

The value behind an attack

xT values progression. xA describes chance creation. xG evaluates the shot.



xT

Expected Threat

How much does a successful pass or carry improve the attacking position? Subtract the start zone value from the end zone value.

0.042 - 0.012 = +0.030

xA

Expected Assists

How good was the chance a pass created? In a shot-based approach, the final pass receives the xG value of the resulting shot.

A 0.30 xG shot gives 0.30 xA

xG

Expected Goals

How likely is a shot to score? Its location, angle and other context inform a probability between zero and one.

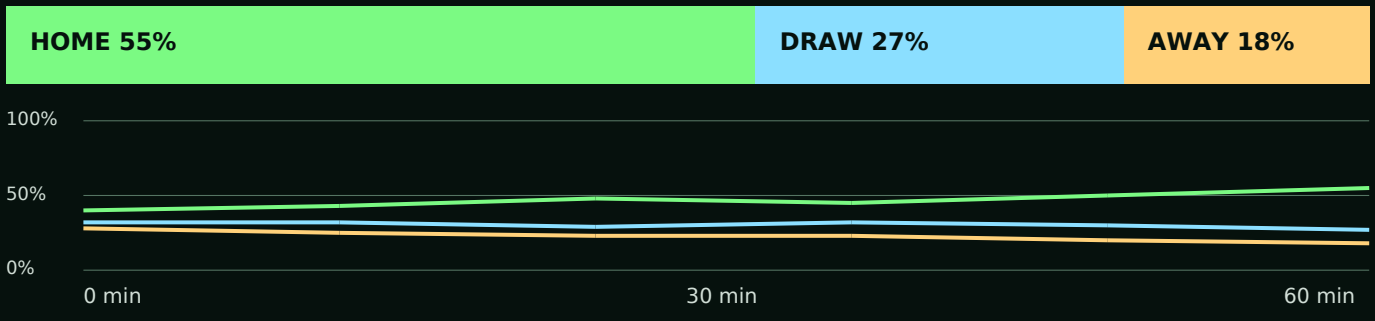
0.30 means about 30 in 100

xA example uses xG assisted. Definitions differ across models.

Live win probability

PROPOSED FEATURE

An evolving estimate of a home win, a draw or an away win as the match progresses. The concept below illustrates the intended view.



All values are illustrative. Probabilities are estimates, not guaranteed outcomes.

A practical starting point

For clubs and analysts, the first conversation starts with your footage and the questions you want it to answer.

A PROPOSED PILOT

01

Your footage

Discuss a suitable match recording and the analysis your team needs.

02

A focused trial

Review tracking, events and selected insights on a defined sample.

03

Shared evaluation

Assess usefulness and reliability, then agree on the next development priorities.

Current product scope

A desktop workflow combining video analysis, tactical drawing overlays and statistical views. Tactical-angle footage with a clear view of play is the intended input.

Ball visibility, tracking continuity and pitch calibration affect results. A pilot should validate the outputs on your footage.

Product direction

Planned development includes explainable player ratings, deeper team-performance analysis and tactical recommendations.

Live win probability extends this direction with an evolving match outlook. It appears here as a proposed capability.

Developers & research collaborators

Contribute to computer vision, football modelling, evaluation or product design. The goal is to strengthen the system around real analysis needs.

THE MATCHLENS TEAM

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